

Health Data Analysis

Dashboard Documentation & Business Questions

Four business-focused dashboards with defined goals, target audiences, KPIs, and SQL business questions

Dashboard	Business Focus
1	Overall Population Health Profile
2	Blood Pressure Risk Across Age Groups
3	Cholesterol and Blood Lipid Risk
4	Lifestyle Factors and Health Outcomes

Dashboard 1: Health Overview

Dashboard Topic

Overall Population Health Profile

Dashboard Goal

Provide decision makers with a high-level view of the population's overall health status and highlight the most important health indicators.

Target Audience

- Healthcare Managers
- Public Health Decision Makers
- Health Program Coordinators
- Senior Management

KPI Requirements

KPI	Purpose
Total Number of People	Shows the size of the analyzed population
Average BMI	Summarizes the overall BMI level
Average Systolic Blood Pressure	Summarizes the average upper blood pressure value
Average Diastolic Blood Pressure	Summarizes the average lower blood pressure value
Average Total Cholesterol	Summarizes the overall cholesterol level

Business Questions

#	Difficulty	Business Question
1	Easy	How many people are included in the dataset?
2	Easy	How many people are smokers and how many are non-smokers?
3	Easy	How many people belong to each physical activity level?
4	Medium	How are people distributed across BMI categories such as Underweight, Normal, Overweight, and Obese?
5	Medium	How many people have desirable total cholesterol and how many have high total cholesterol?
6	Medium	How many people have normal systolic blood pressure and how many have elevated or high systolic blood pressure?
7	Hard	What percentage of the population falls into each BMI-based health category?

Question mix: 3 Easy questions, 3 Medium questions, and 1 Hard question. KPIs are separate from the seven SQL business questions.

Dashboard 2: Blood Pressure & Age

Dashboard Topic

Blood Pressure Risk Across Age Groups

Dashboard Goal

Identify age groups with higher blood pressure risk and show how blood pressure status changes across different stages of age.

Target Audience

- Clinical Risk Analysts
- Healthcare Professionals
- Preventive Care Teams
- Health Risk Assessment Teams

KPI Requirements

KPI	Purpose
Average Systolic Blood Pressure	Summarizes the average upper blood pressure value
Average Diastolic Blood Pressure	Summarizes the average lower blood pressure value
Highest Systolic Blood Pressure	Highlights the maximum observed systolic value
Highest Diastolic Blood Pressure	Highlights the maximum observed diastolic value
Percentage with High Blood Pressure	Shows the share of the population with high blood pressure

Business Questions

#	Difficulty	Business Question
1	Easy	What are the minimum and maximum systolic blood pressure values?
2	Easy	What are the minimum and maximum diastolic blood pressure values?
3	Easy	How many people have systolic blood pressure greater than or equal to 130?
4	Medium	How many people belong to each age group: 18–29, 30–39, 40–49, 50–59, and 60+?
5	Medium	How many people have high blood pressure in each age group?
6	Medium	How many people have normal blood pressure compared with elevated or high blood pressure?
7	Hard	Which age group has the highest percentage of people classified as having high blood pressure?

Question mix: 3 Easy questions, 3 Medium questions, and 1 Hard question. KPIs are separate from the seven SQL business questions.

Dashboard 3: Cholesterol Analysis

Dashboard Topic

Cholesterol and Blood Lipid Risk

Dashboard Goal

Monitor cholesterol and blood lipid indicators to identify groups with potentially higher cardiovascular risk and determine where attention is needed.

Target Audience

- Cardiovascular Risk Analysts
- Clinical Nutrition Teams
- Health Screening Teams
- Healthcare Researchers

KPI Requirements

KPI	Purpose
Average Total Cholesterol	Summarizes the overall cholesterol level
Average HDL Cholesterol	Summarizes the average HDL level
Average LDL Cholesterol	Summarizes the average LDL level
Average Triglycerides	Summarizes the average triglyceride level
Percentage with High LDL	Shows the share of the population with high LDL

Business Questions

#	Difficulty	Business Question
1	Easy	What is the highest total cholesterol value in the dataset?
2	Easy	What is the lowest LDL cholesterol value in the dataset?
3	Easy	How many people have HDL cholesterol below 40?
4	Medium	How many people fall into the Desirable, Borderline High, and High total cholesterol categories?
5	Medium	How many people have LDL cholesterol classified as High or Very High?
6	Medium	How does the number of people with high total cholesterol differ between smokers and non-smokers?
7	Hard	Which age group has the highest percentage of people with high LDL cholesterol?

Question mix: 3 Easy questions, 3 Medium questions, and 1 Hard question. KPIs are separate from the seven SQL business questions.

Dashboard 4: Lifestyle & Health

Dashboard Topic

Lifestyle Factors and Health Outcomes

Dashboard Goal

Explore how lifestyle characteristics such as smoking and physical activity are associated with BMI, LDL cholesterol, and triglyceride levels.

Target Audience

- Lifestyle and Wellness Analysts
- Health Promotion Teams
- Community Health Planners
- Wellness Program Managers

KPI Requirements

KPI	Purpose
Percentage of Smokers	Shows the share of smokers in the population
Percentage with BMI 25 or Higher	Shows the share classified as overweight or above
Average BMI	Summarizes the overall BMI level
Average LDL Cholesterol	Summarizes the average LDL level
Average Triglycerides	Summarizes the average triglyceride level

Business Questions

#	Difficulty	Business Question
1	Easy	How many people are smokers and how many are non-smokers?
2	Easy	How many people belong to Low, Moderate, and High physical activity levels?
3	Easy	How many people have a BMI greater than or equal to 25?
4	Medium	For each physical activity level, how many people are Overweight or Obese?
5	Medium	How does the number of people with high LDL cholesterol differ between smokers and non-smokers?
6	Medium	For each physical activity level, how many people have high triglycerides?
7	Hard	Which physical activity level has the highest percentage of people with BMI greater than or equal to 25?

Question mix: 3 Easy questions, 3 Medium questions, and 1 Hard question. KPIs are separate from the seven SQL business questions.